

Transverse Quenching in Quartet Coefficient Space

Exact Anisotropy Near the Degenerate Axial Boundary

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Abstract

The centered conjugate-quartet manifold has axial coordinate u and asymmetry coordinate η . RH-216 proves that $u \rightarrow 1$ forces the polynomial toward $(z^2 - 1)^2$ uniformly in η . We sharpen that statement by computing the full coefficient Jacobian and proving an exact transverse Lipschitz bound.

For

$$\Phi(u, \eta) = (2 - 4u, 4\sqrt{u}(1 - u)\eta, 1 - \eta^2(1 - u)^2),$$

the η derivative satisfies

$$\sup_{|\eta| \leq 1} \|\partial_\eta \Phi(u, \eta)\|_2 \leq 2(1 - u)\sqrt{4u + (1 - u)^2}.$$

The same constant controls every finite transverse chord. It vanishes linearly as $u \rightarrow 1$. In contrast, $\|\partial_u \Phi\| \geq 4$ because the first component is always -4 . Thus coefficient space becomes intrinsically anisotropic: axial motion remains visible while transverse motion is quenched.

On the sixteen-level physical atlas, the maximum channelwise transverse-to-axial Jacobian ratio decreases from 0.3752 at $\sigma = 0.02$ to 0.2389 at $\sigma = 0.00125$. The finest left/right transverse coefficient leg is only 0.00136. These finite values support the geometric mechanism but do not prove $u \rightarrow 1$, convergence of η , or a one-dimensional physical flow.

1 Why a narrow corridor may become coefficientwise invisible

RH-214 observes that η fluctuates in a mature interval of width below 0.038, while u increases across every sampled transition [3]. RH-216 shows that the entire $u = 1$ edge of the shape rectangle maps to one polynomial [4]. These facts suggest a specific mechanism: the coefficient map may contract transverse differences as the axial coordinate grows.

This paper proves that mechanism directly. It is a statement about the exact quotient geometry of RH-213, independent of any statistical fit or physical limit [2].

2 Coefficient map

Discarding the fixed leading and cubic coefficients, define

$$\Phi(u, \eta) = \begin{pmatrix} c_2 \\ c_3 \\ c_4 \end{pmatrix} = \begin{pmatrix} 2 - 4u \\ 4\sqrt{u}(1 - u)\eta \\ 1 - \eta^2(1 - u)^2 \end{pmatrix}, \quad (1)$$

for $0 < u \leq 1$ and $|\eta| \leq 1$.

Proposition 2.1 (Exact Jacobian). *For $u > 0$,*

$$D\Phi(u, \eta) = \begin{pmatrix} -4 & 0 \\ \frac{2\eta(1 - 3u)}{\sqrt{u}} & 4\sqrt{u}(1 - u) \\ 2\eta^2(1 - u) & -2\eta(1 - u)^2 \end{pmatrix}. \quad (2)$$

Proof. Differentiate the three components of (1). In particular,

$$\frac{d}{du}[\sqrt{u}(1 - u)] = \frac{1 - 3u}{2\sqrt{u}}. \quad (3)$$

□

The singularity in the displayed u derivative at $u = 0$ reflects loss of the positive/negative real branch label there. The physical mature region stays away from that boundary.

3 Uniform transverse bound

Theorem 3.1 (Transverse Lipschitz quenching). *For fixed $u \in [0, 1]$ and any $\eta_1, \eta_2 \in [-1, 1]$,*

$$\|\Phi(u, \eta_1) - \Phi(u, \eta_2)\|_2 \leq L_\eta(u)|\eta_1 - \eta_2|, \quad (4)$$

where

$$\boxed{L_\eta(u) = 2(1 - u)\sqrt{4u + (1 - u)^2}}. \quad (5)$$

Consequently $L_\eta(u) = O(1 - u)$ as $u \rightarrow 1$.

Proof. At fixed u , the coefficient difference is

$$\begin{pmatrix} 0 \\ 4\sqrt{u}(1 - u)(\eta_1 - \eta_2) \\ -(1 - u)^2(\eta_1 + \eta_2)(\eta_1 - \eta_2) \end{pmatrix}. \quad (6)$$

Since $|\eta_1 + \eta_2| \leq 2$, its squared norm is at most

$$4(1 - u)^2[4u + (1 - u)^2]|\eta_1 - \eta_2|^2. \quad (7)$$

Taking square roots proves the estimate. The asymptotic order follows because the square-root factor tends to two. □

This is a global chord estimate, not merely a local derivative statement. It remains valid at $u = 0$ and on the η edges.

4 Axial sensitivity does not vanish

Proposition 4.1 (Persistent axial visibility). *For every $u > 0$ and $|\eta| \leq 1$,*

$$\|\partial_u \Phi(u, \eta)\|_2 \geq 4. \quad (8)$$

Proof. The first component of $\partial_u \Phi$ is -4 . $\square$

Combining Theorem 3.1 and Proposition 4.1 gives the uniform anisotropy ratio

$$\frac{\sup_{\eta} \|\partial_{\eta} \Phi(u, \eta)\|_2}{\inf_{\eta} \|\partial_u \Phi(u, \eta)\|_2} \leq \frac{1-u}{2} \sqrt{4u + (1-u)^2} \rightarrow 0. \quad (9)$$

Thus the map itself, not a fitted trajectory, singles out the axial direction near $u = 1$.

5 Coefficient quenching versus root quenching

The imaginary root heights are

$$b = \sqrt{(1-u)(1+\eta)}, \quad d = \sqrt{(1-u)(1-\eta)}. \quad (10)$$

Their η derivatives can become large near $|\eta| = 1$ because of the square root. Therefore Theorem 3.1 is specifically a coefficient-space theorem. Symmetric polynomials cancel and square the transverse root motion, producing the stronger $O(1-u)$ coefficient scale.

This distinction matters for the divisor-first route. A scalar determinant can stabilize even when individual branch coordinates are less regular, just as RH-210 showed that divisor stability need not imply projector stability [1].

6 Exact path decomposition between channels

At one scale let the left and right shapes be (u_L, η_L) and (u_R, η_R) . Insert the corner (u_R, η_L) :

$$\Phi(u_R, \eta_R) - \Phi(u_L, \eta_L) = \underbrace{\Phi(u_R, \eta_L) - \Phi(u_L, \eta_L)}_{\text{axial leg}} + \underbrace{\Phi(u_R, \eta_R) - \Phi(u_R, \eta_L)}_{\text{transverse leg}}. \quad (11)$$

This is an exact telescoping identity. The transverse leg obeys

$$\|\text{transverse leg}\|_2 \leq L_{\eta}(u_R) |\eta_R - \eta_L|. \quad (12)$$

The frozen audit evaluates both legs at all sixteen scales. Eight hundred random chord tests have no positive bound violation, and all path residuals are machine zero.

7 Finite physical anisotropy

The exact Jacobian is evaluated at each of the 32 physical shape points. Representative mature values are summarized by

diagnostic	$\sigma = 0.02$	$\sigma = 0.00125$
maximum transverse/axial Jacobian ratio	0.37516	0.23883
maximum channel $ \Delta\eta $	0.00145	0.00141
maximum transverse channel-leg norm	small	0.001354

The ratio decreases as the observed axial clock advances, in agreement with (9). It is not yet extremely small: $u \approx 0.718$ remains far from the degenerate edge. The numbers therefore support the mechanism without demonstrating an asymptotic regime.

8 Implications and limitations

If a future theorem proves $u_\sigma \rightarrow 1$, then any bounded η_σ sequence—convergent or not—has vanishing transverse coefficient effect. The coefficient path could become asymptotically one-dimensional even while the transverse coordinate continues to oscillate.

The converse is not supplied here. Finite decline of the sensitivity ratio does not prove $u \rightarrow 1$, and coefficient anisotropy does not identify a scale-independent map advancing u . RH-218 tests that stronger recurrence claim separately.

No growing root cloud, locally uniform determinant, Gate-A closure, Hilbert–Pólya operator, arithmetic trace formula, or zeta identification is obtained.

References

- [1] Liang Wang. A divisor-first pivot inside gate a, 2026. RH-210 preprint.
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